SHRIKRISHNA EDUCATIONAL AND CULTURAL MANDAL'S

# SHRI GULABRAO DEOKAR POLYTECHNIC,

Gat no. 26, Mohadi shivar, Shirsoli road, Jalgaon, Pin: 425002 (M.S)

## A Capstone Project report on

"Air Quality Prediction"

**Is Submitted as per I scheme Curriculum and the requirement for the Program: Diploma Third Year in Computer Engineering.**

**Group Members:**

| Sr .no | Name of members | Enrollment number | Roll no. |
|---|---|---|---|
| 1. | Atharva Kundan Rane | 2005090068 | 25 |
| 2. | Tejas Anil Rane | 2005090059 | 17 |
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**Under the Guidance of:**
**Prof. Mrs. M.V.Yeole**

**Affiliated to**

**MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION MUMBAI.51**

**Academic year 2022-23**

SHRIKRISHNA EDUCATIONAL AND CULTURAL MANDAL'S

SHRI GULABRAO DEOKAR POLYTECHNIC, JALGAON

DEPARTMENT OF COMPUTER ENGINEERING

# CERTIFICATE

**This is to verify that**

| Sr.no | Name of members | Enrollment number | Roll no. |
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| 1. | Atharva Kundan Rane | 2005090068 | 25 |
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**Have successfully presented the entitled capstone project on "Air Quality Prediction" and submitted in satisfactory manner.**

**The major project is submitted in partial fulfillment for the third year diploma in computer engineering affiliated to Maharashtra state board of technical education, Mumbai for academic year 2022-23**

| | | |
|---|---|---|
| Prof.Mrs.M.V. Yeole<br>(Guide) | Prof.Dr.M.P.Patil<br>(Principal) | Prof. Mrs.M.V. Yeole<br>(H.O.D.) |

# Acknowledgement

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Lastly, our cordial thanks to all who have contributed indirectly and materially in words and deeds for completion of Capstone project planning.

# ABSTRACT

The Air Quality Index (AQI) is an index that records daily air quality. It is a measure of how air pollution affects one's health in a short period of time. AQI aims to inform people about how local air quality affects their health. The Environmental Protection Agency (EPA) calculates AQI for five major air pollutants, for which national air quality standards have been established to protect public health. The project is an important impetus for determining and sustaining the city's air quality index. For today, tonight and tomorrow, AQI provides local air quality value and local air quality maxima forecasts and provides relevant health advice to people living in that particular area who may be useful for a particular occasion. Air emissions increase during rush hour, during traffic, or when forest fires increase air pollutants . This allows air pollution to remain in the local area, leading to higher levels of pollutants, chemical reactions that occur in conditions such as air pollution and fumes. The project aims to help people learn how to measure the air quality index and how the local climate affects health and lifestyle.

Key words: SVM, ML, Training Module, dataset.

# List of Abbreviations

The next list describes several abbreviations that will be later used within the body of the document

*ANN* Convolutional Neural Network

*ANN* K-Nearest Neighbor

*SVM* Support Vector Machine

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# INTRODUCTION

AQI is an index that records daily air quality. AQI is calculated on a scale of 0 to 500. The higher the AQI value, the higher the level of air pollution and the higher the health concern. For example, an AQII value of 50 indicates good air quality that is unlikely to affect public health, while an AQII value greater than 300 represents hazardous air quality. The AQII value of 100 generally corresponds to the national air quality standards of pollutants, which is the level of the Central Pollution Control Board to protect public health. AQI values below 100 are generally considered satisfactory. When AQI values are above 100, air quality is considered harmful to health - first for some sensitive groups of people, then higher for everyone. AQI considers the main pollutants - PM (Rough: PM-10, Fine: PM-2.5), $NO_2$, $SO_2$, CO, $O_3$, $NH_3$ and PB - for a short period of time (up to an average of 24 hours) of national ambient air quality. Standards are prescribed.

PM or Particulate Matter is a mixture of solids and liquid drops floating freely in the air. Some particles are released directly by specific sources, while others undergo complex chemical reactions in the atmosphere. Particles come in different sizes. Particles less than or equal to 10 micrometres in diameter are so small that they can go into the lungs and potentially cause serious health problems.
The sub-index for each of these pollutants is calculated based on the measured atmospheric concentrations, the relevant standards and the potential health effects. The worst sub-index reflects the overall AQI. The potential health effects associated with different AQI categories and pollutants have also been suggested with preliminary information from medical expert members of this group. The AQI values and the associated environmental concentrations (health breakpoints) as well as the associated health effects for the eight identified pollutants are as follows:
Good: (0-50)Minimal effect
Satisfactory (51-100)sensitive people may find it difficult to breathe.
People with moderately polluted (101-200) lung diseases such as asthma can suffer from respiratory distress and children with heart disease, infants and the elderly. Poor(201-300) people with long-term risk may have difficulty breathing and people with heart disease may have difficulty.

Very poor (301-400) people with long-term risk can get respiratory illness. Its effects are more pronounced in people with lung and heart disease.

Severe (401-500) respiratory effects can also affect normal healthy people and people with lung / heart disease can have serious health effects. Health effects can also be felt during light physical activity.

# Problem Statement

To Determining the Air Quality Index of the city for the standard living using Machine Learning Algorithm.

## Scope

Useful for predicting the air quality with the quantity of pollutant present in the air.

## Objective

To develop an efficient and effective model which predicts the air quality according to the user's inputs.

To achieve good accuracy.

To develop a User Interface( UI ) which is user-friendly and takes input from the user and predicts the price

# Chapter 2:

# LITERATURE REVIEW

Some air pollutants can be dangerous as taking them inside the body can lead to serious health problems [1]. It can harm anyone, young or old, most especially those people with lung diseases. According to the World Health Organization (WHO), there are about three million deaths every year as a result of exposure to ambient or outdoor air pollution. Around 4.3 million deaths every year due to the household exposure to smoke from dirty cookstoves and fuels and about 92% of the world's population lives in places where air quality exceeds WHO guideline limits [2].

One way of providing public awareness is to enable the people living in a certain place or area to know the air quality that they are inhaling and to be informed on the possible health problems that they might experience. It is important to know that the air you are taking in is clean and people would only understand it by knowing the AQI. AQI provides information on how clean the air is. The lower the AQI, the better the quality of the air [3]. AQI is used to report the day to day air quality with regard to human health and the environment [4]. This study focuses on determining the value of AQI using fuzzy logic (FL).FL in a narrow sense, is a logical system type that can be regarded as an extension of multivalued logic. Taking it in a wider sense, FL can be considered as a theory that relates to the classes of objects with not sharp boundaries using some degree of membership functions [5]. Using the set of user-input language rules, it is where FL starts. By converting these rules into their mathematical equivalents, fuzzy systems enable to simplify the work of the system designer which can give more accurate results on how the system behaves in the real world [6]. Among its benefits are simplicity and flexibility. Even with incomplete and imprecise data, FL can resolve that problem and model nonlinear functions for which the complexity is arbitrary.

With the help of the fuzzy logic toolbox, matching any set of input-output data can be made easy by utilising adaptive techniques such as fuzzy subtractive clustering and adaptive neuro-fuzzy inference systems (ANFIS) [6].
Determining the quality of its air is a very important task for the government as they can give warnings to their people or residents in their area. By determining their respective AQI, the local governments can take the necessary actions on how to keep their air safe for their people.

# Chapter 3:

# Software Requirement Specification

In the previous chapter, we saw the Literature Survey conducted on various reference papers for the project. This survey helped us frame our problem statements and objectives. It also helped us in selecting our tools and methodologies for the project.

The requirement specifications of any project are the most important while starting one. This chapter will be discussing all the specifications required for this project. It discusses all the functional, non-functional requirements, external interface requirements, system requirements, etc. for the project.

## System Requirements

### Software Requirements:

1. Pycharm 2020.2.3
2. Python 2.7

### Hardware Requirements:

1. Operating system: Windows 10
2. Processor - Intel i3/i5/i7
3. Speed -1.1 GHz
4. RAM - 8 GB(min)
5. Hard Disk - 500 GB
6. Key Board - Standard Windows Keyboard
7. Mouse - Two or Three Button Mouse

# Non Functional Requirements

Non-functional requirements are requirements which specify criteria that can be used to judge the operation of a system, rather than specific behaviors. This should be contrasted with functional requirements that specify specific behavior or functions. Typical non- functional requirements are reliability, scalability, and cost. Other terms for non-functional requirements are ”constraints”, ”quality attributes” and ”quality of service requirements”.

1. Reliability

   If any exceptions occur during the execution of the software it should be caught and there by prevent the system from crashing.

2. Scalability

   The system should be developed in such a way that new modules and functionalities can be added, thereby facilitating system evolution.

3. Accuracy

   System should correctly execute process; display the result i.e. exact detection of disease. System should be able to suggest accurate pesticides.

## Performance Requirements

- The performance of the functions and every module must be well.
- The overall performance of the software will enable the users to work efficiently.

## Software Quality Attributes

- The software is user friendly while using it.

# Chapter 4:

# Methodology:

Methodologies:

- Data understanding and exploration
- Data cleaning
- Data preparation
- Model building and evaluation

**1. Data Preprocessing:**

"Before Training, any model using any algorithm Data Preprocessing is that the most significant step and will be the primary step. the data Preprocessing contains several checkpoints (steps) such as: "10

1. Step 1:

Import Libraries: The essential Libraries for Data preprocessing I used are Pandas for data manipulation and analysis, Numpy for numerical analysis, Matplotlib and Seaborn for better visuals and graphical stats of the data.

2. Step 2:

Import the Dataset: This downloaded this dataset from Kaggle, and then downloaded the dataset using the pandas library.

3. Step 3:

Taking care of Missing Data in Dataset: After evaluation of this dataset, I found no missing values in the dataset.

4. Step 4:

Encoding categorical data: This dataset contains some Categorical values such as fuel type, owner type, seller type, so we need to encode these categorical data into an encoded format to better train our model, to do this I used get_Dummies() method of pandas and this converted the whole Categorical values in the dataset into binary values.

5. Step 5:

Splitting the Dataset into the Training set and Test Set: To split this dataset into Test and Train dataset to train our machine learning model I used the capable machine learning library of python, scikit-learn or sklearn. Using its model selection method to create testing data by picking random values from the available dataset for model prediction, or we can say Supervised Learning.

6. Step 6:
Feature Scaling: Since all the data, available in a standard format, so here I do not use any feature scaling techniques.

**2. Data Training and Modelling**:

To train and develop a model, first of all, we need to the dependent and independent variables. To find these variables, first I used to find the correlation between the variables of the output and then separates my variables into two different axes we call it x and y where the x-axis contains all the independent variable and y-axis having the dependent variable, in our model its selling price of the Used Cars. Using sklearn.model_selection library and its train_test_split function, further this dataset is distributed in the train-test dataset using RandomizedSearchCV tuning of this model is done to find the best hyperparameters for our model prediction.

# Chapter 5: Details of design and working process

## Analysis Models:

**SDLC Model to be applied**

Software Development Life Cycle (SDLC) is a framework that defines the steps involved in the development of software at each phase. It covers the detailed plan for building, deploying and maintaining the software. SDLC defines the complete cycle of development i.e. all the tasks involved in planning, creating, testing, and deploying a Software Product. Purpose of SDLC is to deliver a high-quality product which is as per the customer's requirement. SDLC has defined its phases as, Requirement gathering, Designing, Coding, Testing, and Maintenance. It is important to adhere to the phases to provide the Product in a systematic manner.

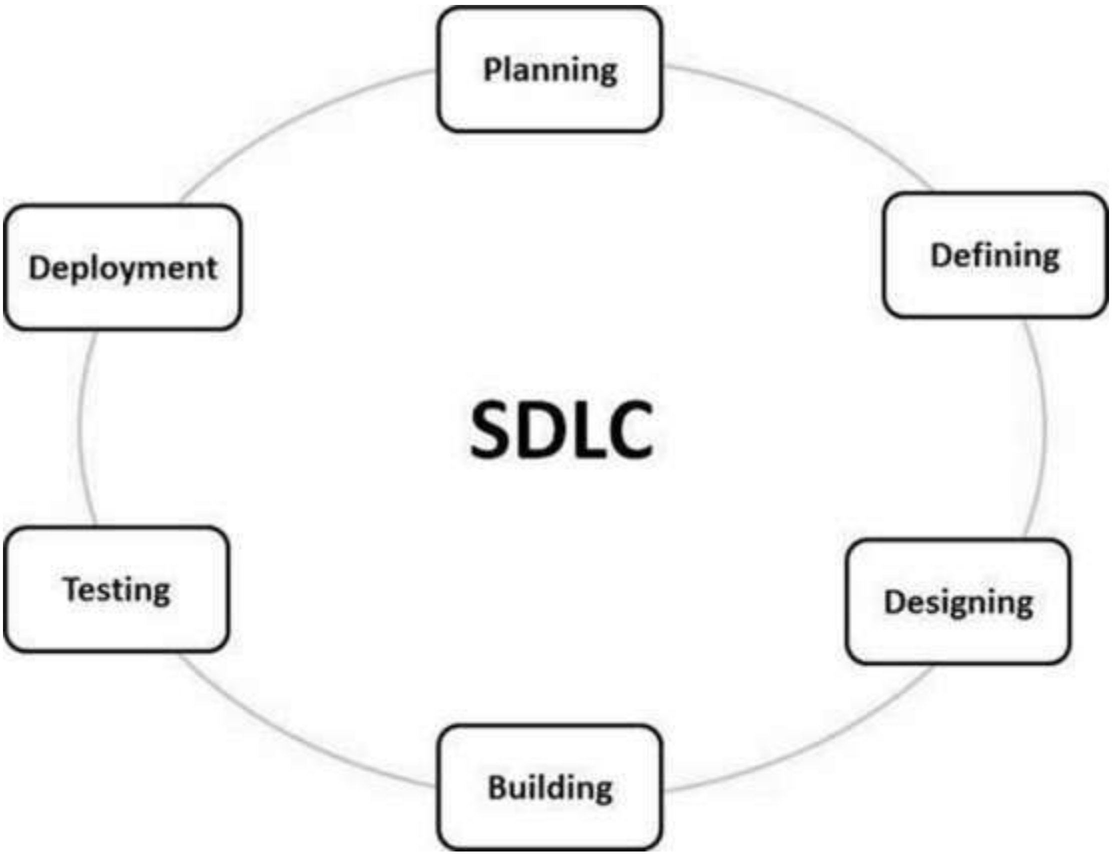


Figure 1 SDLC

**1) Requirement Gathering and Analysis:**

During this phase, all the relevant information is collected from the customer to develop a product as per their expectation. Any ambiguities must be resolved in this phase only.

Business analyst and Project Manager set up a meeting with the customer to gather all the information like what the customer wants to build, who will be the end-user, what is the purpose of the product. Before building a product a core understanding or knowledge of the product is very important.

**For Example,** A customer wants to have an application which involves money transactions. In this case, the requirement has to be clear like what kind of transactions will be done, how it will be done, in which currency it will be done, etc.

Once the requirement gathering is done, an analysis is done to check the feasibility of the development of a product. In case of any ambiguity, a call is set up for further discussion.

Once the requirement is clearly understood, the SRS (Software Requirement Specification) document is created. This document should be thoroughly understood by the developers and also should be reviewed by the customer for future reference.

## 2) Design:

In this phase, the requirement gathered in the SRS document is used as an input and software architecture that is used for implementing system development is derived.

## 3) Implementation or Coding:

Implementation/Coding starts once the developer gets the Design document. The Software design is translated into source code. All the components of the software are implemented in this phase.

## 4) Testing:

Testing starts once the coding is complete and the modules are released for testing. In this phase, the developed software is tested thoroughly and any defects found are assigned to developers to get them fixed.

Retesting, regression testing is done until the point at which the software is as per the customer's expectation. Testers refer SRS document to make sure that the software is as per the customer's standard.

## 5) Deployment:

Once the product is tested, it is deployed in the production environment or first_ is done depending on the customer expectation.

In the case of UAT, a replica of the production environment is created and the customer along with the developers does the testing. If the customer finds the application as expected, then sign off is provided by the customer to go live.

## 6) Maintenance:

After the deployment of a product on the production environment, maintenance of the product i.e. if any issue comes up and needs to be fixed or any enhancement is to be done is taken care by the developers.

A software life cycle model is a descriptive representation of the software development cycle. SDLC models might have a different approach but the basic phases and activity remain the same for all the models.

## Waterfall model :

It is the very first model that is used in SDLC. It is also known as the linear sequential model.

In this model, the outcome of one phase is the input for the next phase. Development of the next phase starts only when the previous phase is complete.

- First, Requirement gathering and analysis is done. Once the requirement is freeze then only the System Design can start. Herein, the SRS document created is the output for the Requirement phase and it acts as an input for the System Design.
- In System Design Software architecture and Design, documents which act as an input for the next phase are created i.e. Implementation and coding.
- In the Implementation phase, coding is done and the software developed is the input for the next phase i.e. testing.
- In the testing phase, the developed code is tested thoroughly to detect the defects in the software. Defects are logged into the defect tracking tool and are retested once fixed. Bug logging, Retest, Regression testing goes on until the time the software is in go-live state.
- In the Deployment phase, the developed code is moved into production after the sign off is given by the customer.

# Components of Software:

There are three components of the software: These are : Program, Documentation, and Operating Procedures.

## Program –

A computer program is a list of instructions that tell a computer what to do.

## Documentation –

Source information about the product contained in design documents, detailed code comments, etc.

# Operating Procedures –

Set of step-by-step instructions compiled by an organization to help workers carry out complex routine operations.
**Code**: the instructions that a computer executes in order to perform a specific task or set of tasks.
**Data**: the information that the software uses or manipulates.
**User** interface: the means by which the user interacts with the software, such as buttons, menus, and text fields.
**Libraries**: pre-written code that can be reused by the software to perform common tasks.
**Documentation**: information that explains how to use and maintain the software, such as user manuals and technical guides.
**Test cases**: a set of inputs, execution conditions, and expected outputs that are used to test the software for correctness and reliability.
**Configuration files**: files that contain settings and parameters that are used to configure the software to run in a specific environment.
**Build and deployment** scripts: scripts or tools that are used to build, package, and deploy the software to different environments.
**Metadata**: information about the software, such as version numbers, authors, and copyright information.
All these components are important for software development, testing and deployment.
There are four basic key process activities:
**Software Specifications –**

In this process, detailed description of a software system to be developed with its functional and non-functional requirements.

**Software Development –**

In this process, designing, programming, documenting, testing, and bug fixing is done.

**Software Validation –**

In this process, evaluation software product is done to ensure that the software meets the business requirements as well as the end users needs.

**Software Evolution –**

It is a process of developing software initially, then timely updating it for various reasons.
**Software Crisis :**

**Size and Cost –**

Day to day growing complexity and expectation out of software. Software are more expensive and more complex.

**Quality –**

Software products must have good quality.

**Delayed Delivery –**

Software takes longer than the estimated time to develop, which in turn leads to cost shooting up.
The term "software crisis" refers to a set of problems that were faced by the software industry in the 1960s and 1970s, such as:
High costs and long development times: software projects were taking much longer and costing much more than expected.
**Low quality**: software was often delivered late, with bugs and other defects that made it difficult to use.
**Lack of standardization**: there were no established best practices or standards for software development, making it difficult to compare and improve different approaches.
**Lack of tools and methodologies**: there were few tools and methodologies available to help with software development, making it a difficult and time-consuming process.
These problems led to a growing realization that the traditional approaches to software development were not effective and needed to be improved. This led to the development of new software development methodologies, such as the Waterfall and Agile methodologies, as well as the creation of new tools and technologies to support software development.
However, even today, software crisis could be seen in some form or the other, like

for example software projects going over budget, schedule and not meeting the requirement.

# Software Process Model

A software process model is an abstraction of the actual process, which is being described. It can also be defined as a simplified representation of a software process. Each model represents a process from a specific perspective. Basic software process models on which different type of software process models can be implemented:

**A workflow Model –**

It is the sequential series of tasks and decisions that make up a business process.

**The Waterfall Model –**

It is a sequential design process in which progress is seen as flowing steadily downwards. Phases in waterfall model:
**(i)** Requirements Specification
**(ii)** Software Design
**(iii)** Implementation
**(iv)**Testing

**Dataflow Model –**

It is diagrammatic representation of the flow and exchange of information within a system.

**Evolutionary Development Model –**

Following activities are considered in this method:
**(i)** Specification
**(ii)** Development
**(iii)**Validation

**Role / Action Model –**

Roles of the people involved in the software process and the activities.

**Need for Process Model:**

The software development team must decide the process model that is to be used for software product development and then the entire team must adhere to it. This is necessary because the software product development can then be done systematically. Each team member will understand what is the next activity and how to do it. Thus process model will bring the definiteness and discipline in overall development process. Every process model consists of definite entry and exit criteria for each phase. Hence the transition of the product through various phases is definite.

If the process model is not followed for software development then any team member can perform any software development activity, this will ultimately cause a chaos and software project will definitely fail without using process model, it is difficult to monitor the progress of software product. Thus process model plays an important rule in software engineering.

**Advantages or Disadvantages:**

There are several advantages and disadvantages to different software development methodologies, such as:

**Waterfall:**

**Advantages:**

Clear and defined phases of development make it easy to plan and manage the project.

It is well-suited for projects with well-defined and unchanging requirements.

**Disadvantages:**

Changes made to the requirements during the development phase can be costly and time-consuming.

It can be difficult to know how long each phase will take, making it difficult to estimate the overall time and cost of the project.

It does not have much room for iteration and feedback throughout the development process.

**Agile:**

**Advantages:**

Flexible and adaptable to changing requirements.

Emphasizes rapid prototyping and continuous delivery, which can help to identify and fix problems early on.

Encourages collaboration and communication between development teams and stakeholders.

**Disadvantages:**

It may be difficult to plan and manage a project using Agile methodologies, as requirements and deliverables are not always well-defined in advance.

It can be difficult to estimate the overall time and cost of a project, as the process is iterative and changes are made throughout the development.

**Scrum:**

**Advantages:**

Encourages teamwork and collaboration.

Provides a flexible and adaptive framework for planning and managing software development projects.

Helps to identify and fix problems early on by using frequent testing and inspection.

**Disadvantages:**

A lack of understanding of Scrum methodologies can lead to confusion and inefficiency.

It can be difficult to estimate the overall time and cost of a project, as the process is iterative and changes are made throughout the development.

**DevOps:**

**Advantages:**

Improves collaboration and communication between development and operations teams.

Automates software delivery process, making it faster and more efficient.

Enables faster recovery and response time in case of issues.

**Disadvantages:**
Requires a significant investment in tools and technologies.
Can be difficult to implement in organizations with existing silos and lack of culture of collaboration.
Need to have a skilled workforce to effectively implement the devops practices.
Ultimately, the choice of which methodology to use depends on the specific project and organization, as well as the goals and requirements of the project.

# PROJECT PLAN

## Reconciled Estimate:

We have considered three main factors of Software Costing while estimating the cost of this project.
**Type of Software Project: -** We are developing a New Software, so this software is neither a type of modification software nor integration software.
**Size of Software Project**: - This is a small scaled software project, with limited requirements and resources.
**Development Team Size: -** Development team size is four. All four members will equally contribute to all the phases of software development.

## Project Resources:

### Software resources:-

Pycharm.
OS (Windows 10)

### Hardware resources:-

Laptop with minimum 8 GB RAM (16GB recommended) and Intel i5 processor or above generation processor.
Android Device with android.

## Risk Management:

## Operational Risk

Algorithms used will be of latest technologies so the phones with old or outdated android version may not support the app.

We are not going for cross platform or hybrid app development, so that might affect total no of users.

The accuracy of algorithm used is not 100% so there may be chances that algorithm may not recognize object correctly.

## Technical Risks

Multiple module implementation

The product is complex to implement, it involves multiple module integrations. So it might be challenging to integrate them. If it will not integrate properly, then there will be the risk of project failure.

## Schedule Risks

Tight schedule

Working on project and also maintaining academics is difficult.

Amidst of exams, placement activities and self-development activities it’s difficult to work on project by sticking to the schedule and deliver it on time, hence as students tight schedule is one of the major risks.

# PROJECT SCHEDULE

**1.** Project Task Set:

| **Task no** | **Task Performed** | **Description** |
|---|---|---|
| **1** | Project Scoping | Determines overall project scope. |
| **2** | Preliminary concept planning | Establishes the team member's ability to undertake work implied by project scope. |
| **3** | Technology Risk Assessment | Evaluates risk associated with the system or application to be implemented. |
| **4** | Proof of concept | Demonstrates the usefulness and viability of a system or application in the software context. |
| **5** | Concept Implementation | Implements the concept representation in a manner that can be analyze by guide and reviewed by customer and is used for social purposes when a concept must be given to other customers. |
| **6** | Approved from Guide | Concept solicits feedback on a new system or application concept from guide. |

*Table 1: Project Task Set*

## 2. Task Network:

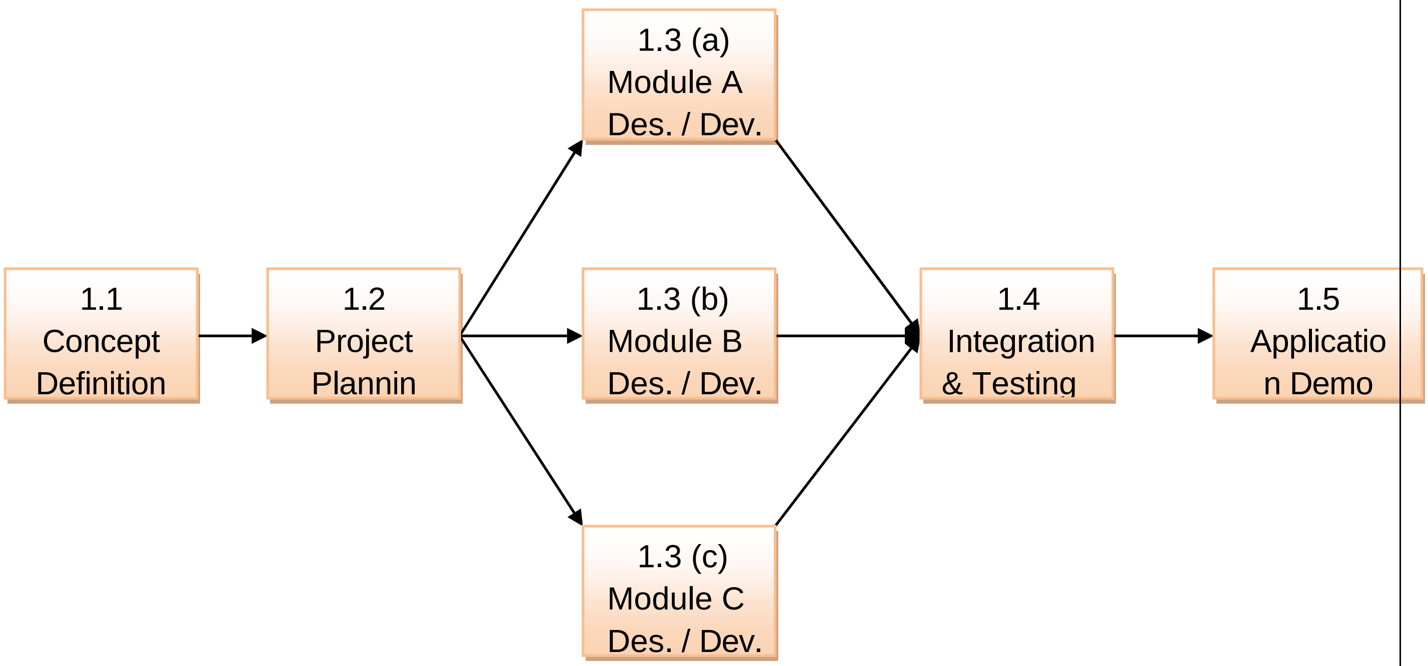


*Figure 2: Task Network*

## 3. Timeline Chart:

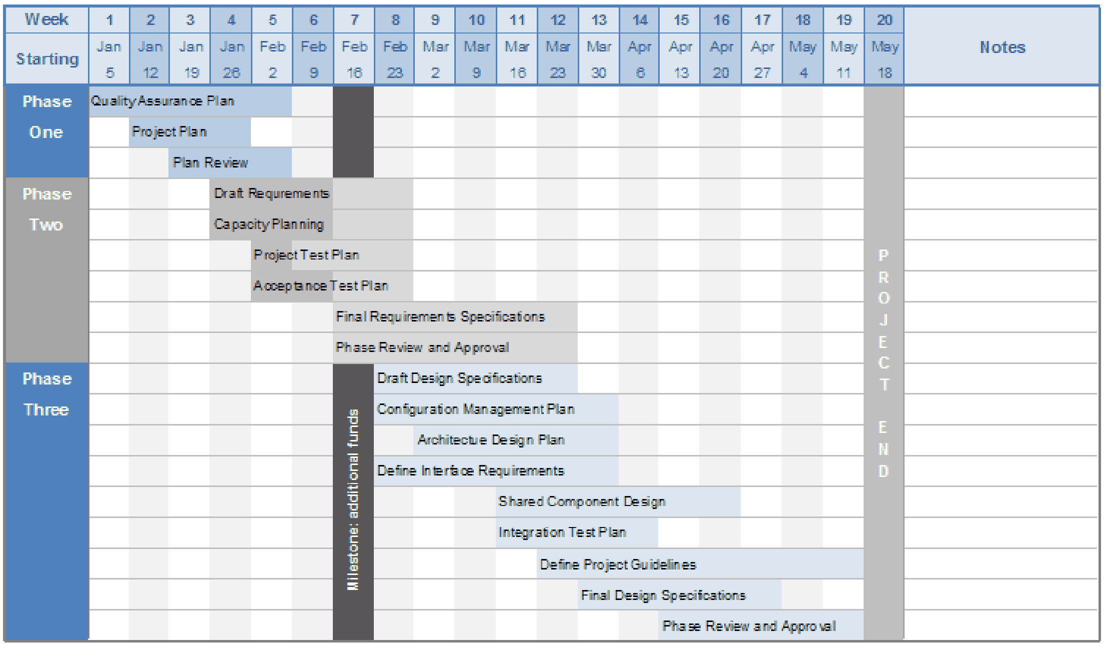


*Figure 3: Timeline Chart*

## TEAM ORGANIZATION

3. Team Structure:

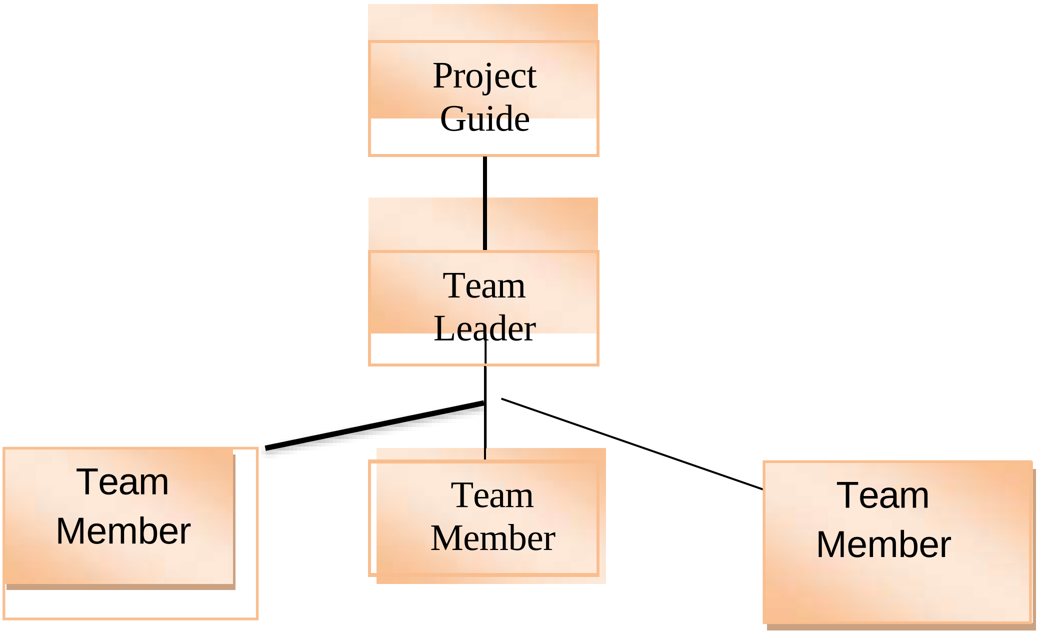


*Figure 4: Team Structure*

2. Management Reporting and Communication:

| Description | Frequency | Method | Audience | Owner |
|---|---|---|---|---|
| **Name of the communication** | **How often it will happen?** | **Method of communication** | **Who will receive the communication?** | **Who is responsible?** |
| Project Team Meeting | Weekly | Virtual Meeting | Project Team | Project Leader |
| Project Guide Update | Weekly | Virtual Meeting | Project Guide | Project Team |
| Project Review Demonstration | Monthly | Virtual Meeting | Project Guide | Project Team |
| Project Development Demonstration | Quarterly | Virtual Meeting | Project Guide | Project Team |

*Table 2: Management Reporting and Communication*

# Algorithms

A. **SUPPORT VECTOR MACHINE(SVM)** SVM is a supervised machine learning algorithm which can be used for both classification or regression challenges. However, it is mostly used in classification problems. In the svm algorithm, we plot each data item as a point in ndimensional space (where n is number of features you have) with the value of each feature being the value of a particular coordinate. Support vector machine is highly preferred by many as it produces significant accuracy with less computation power. SVM can be used for both regression and classification tasks. But, it is widely used in classification objectives. The objective is to look for the optimal separating hyperplane between classes. The points lying on classes' boundaries are called support vectors, and the in-between space is called the hyperplane, when a linear separator is not able to find a solution, data points are projected into a higher-dimensional space, where the previous nonlinearly separable points become linearly separable, using kernel functions. B.

**K-NEAREST NEIGHBORS(KNN)**: In pattern recognition, the k-nearest neighbors algorithm is a non-parametric method used for classification and regression. In both cases, the input consists of the k closest training examples in the feature space. K-Nearest Neighbour is one of the simplest Machine Learning algorithms based on supervised learning technique. K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories. K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm. K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the classification problems. K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data. It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset. KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

**Convolutional Neural N.etwork(CNN)**: Convolutional Neural Network (ConvNet/CNN) is a Deep Learning algorithm which can take in an input image, assign importance (learnable weights and biases) to various aspects/objects in the image and be able to differentiate one from the other. The preprocessing required in a ConvNet is much lower as compared to other classification algorithms. While in primitive methods filters are hand-engineered, with enough training, ConvNets have the ability to learn these filters/characteristics. The architecture of a ConvNet is analogous to that of the connectivity pattern of Neurons in the Human Brain and was inspired by the organization of the Visual Cortex. Individual neurons respond to stimuli only in a restricted region of the visual field known as the Receptive Field. A collection of such fields overlap to cover the entire visual area

# System Architecture

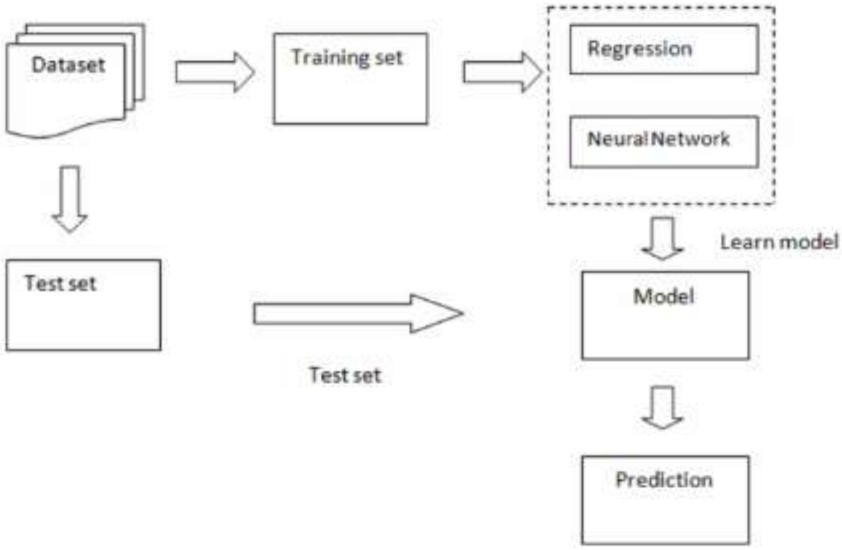


Figure 5 System Architecture

## DATASET DETAILS

- This data is a cleaner version of the Historical Daily Ambient Air Quality Data released by the Ministry of Environment and Forests and Central Pollution Control Board of India under the National Data Sharing and Accessibility Policy (NDSAP).
- The dataset contains the following features :
- stn_code : Station code. A code is given to each station that recorded the data.
- sampling_date: The date when the data was recorded.
- state: It represents the states whose air quality data is measured.
- location: It represents the city whose air quality data is measured.
- agency: Name of the agency that measured the data.
- type: The type of area where the measurement was made.
- so2: The amount of Sulphur Dioxide measured.
- no2: The amount of Nitrogen Dioxide measured
- rspm: Respirable Suspended Particulate Matter measured.
- spm: Suspended Particulate Matter measured.
- location_monitoring_station: It indicates the location of the monitoring area.
- pm2_5: It represents the value of particulate matter measured.
- date: It represents the date of recording (It is a cleaner version of 'sampling_date' feature)

# Data Flow Diagrams:

Data Flow Diagram – Level 0:

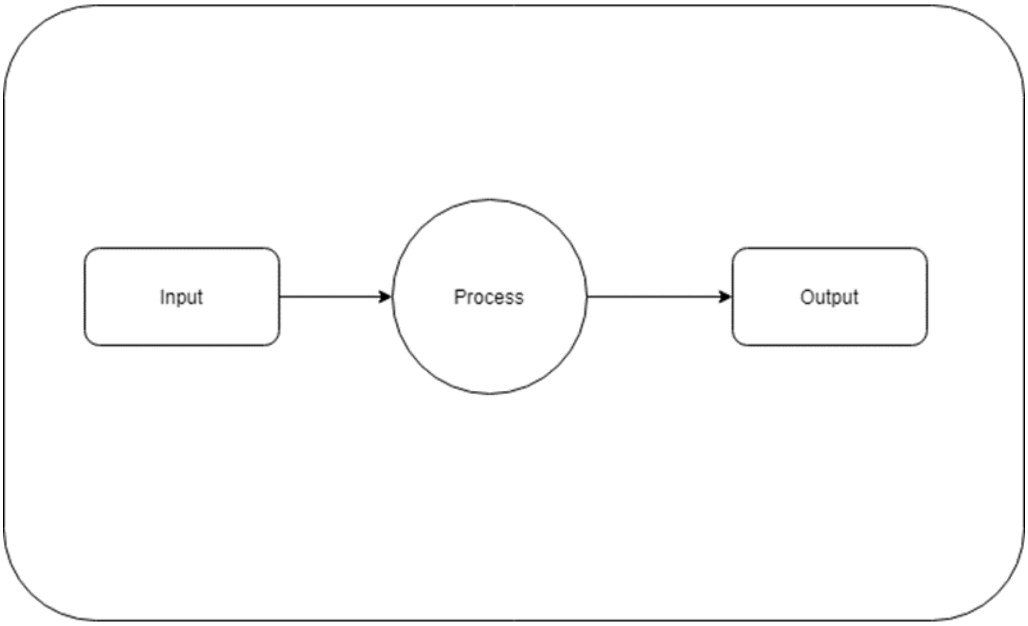


*Figure 6: Data Flow Diagram – Level0*

# UML Diagrams

## Flow Diagram

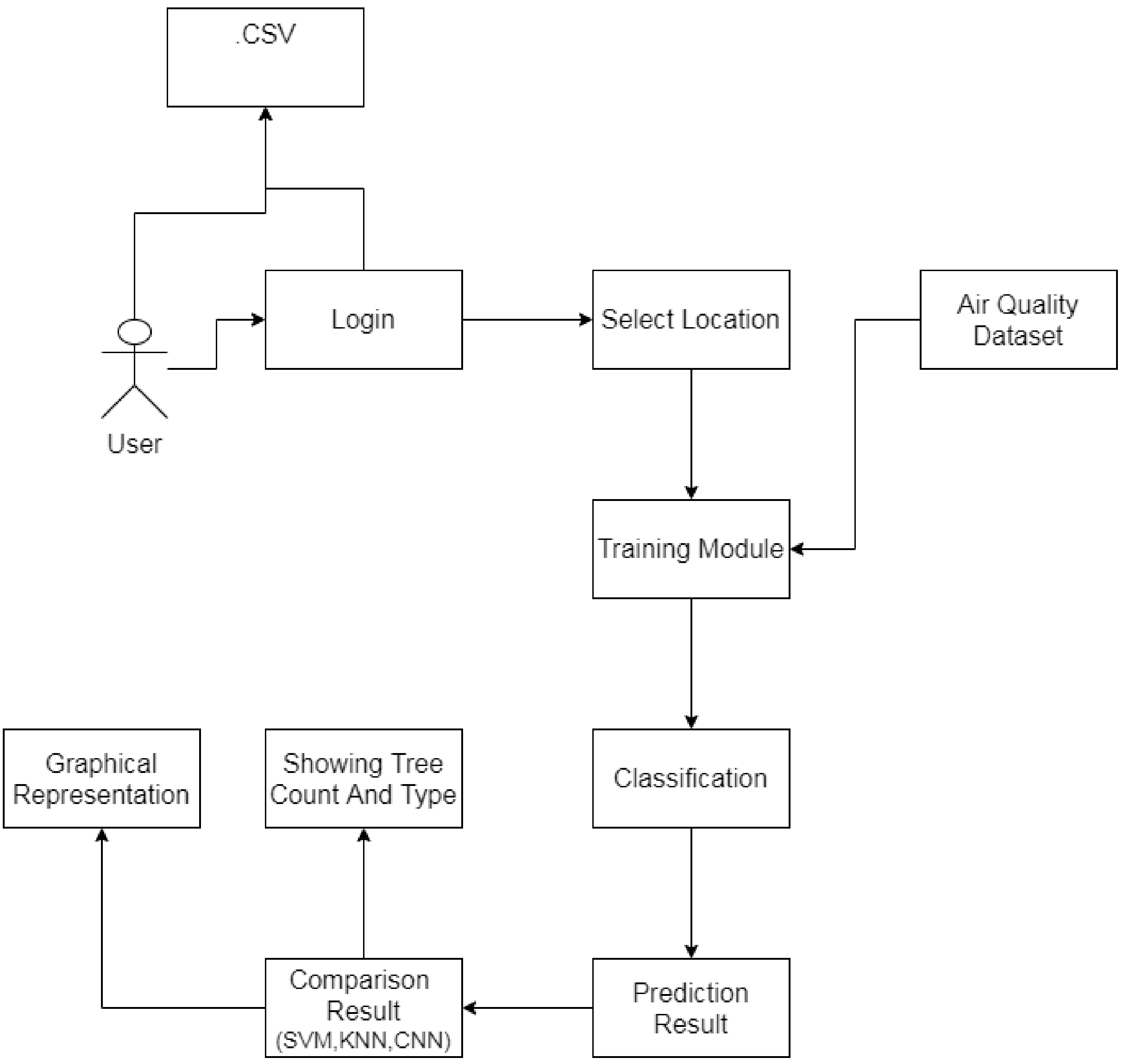


Figure 7.1 Flow Diagram

## Use Case Diagram

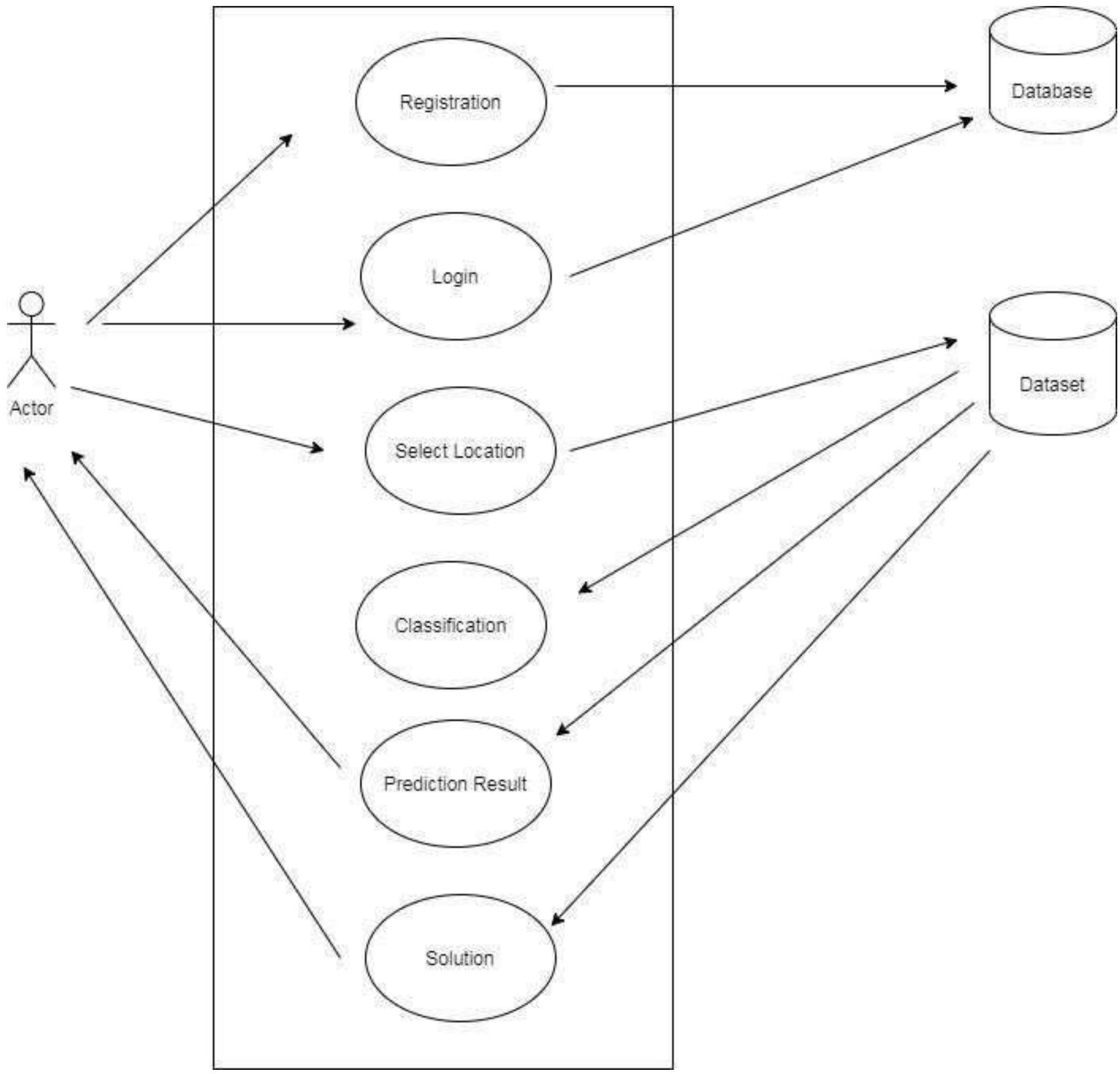


Fig.7.2. Use Case Diagram

Sequence Diagram

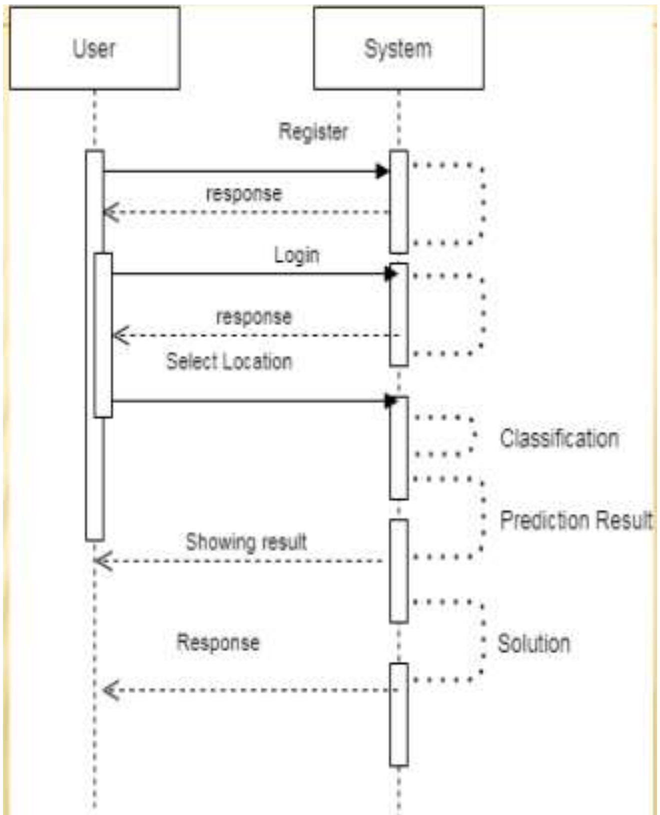


Fig.7.3. Sequence Diagram

# Chapter 6:Result and Application

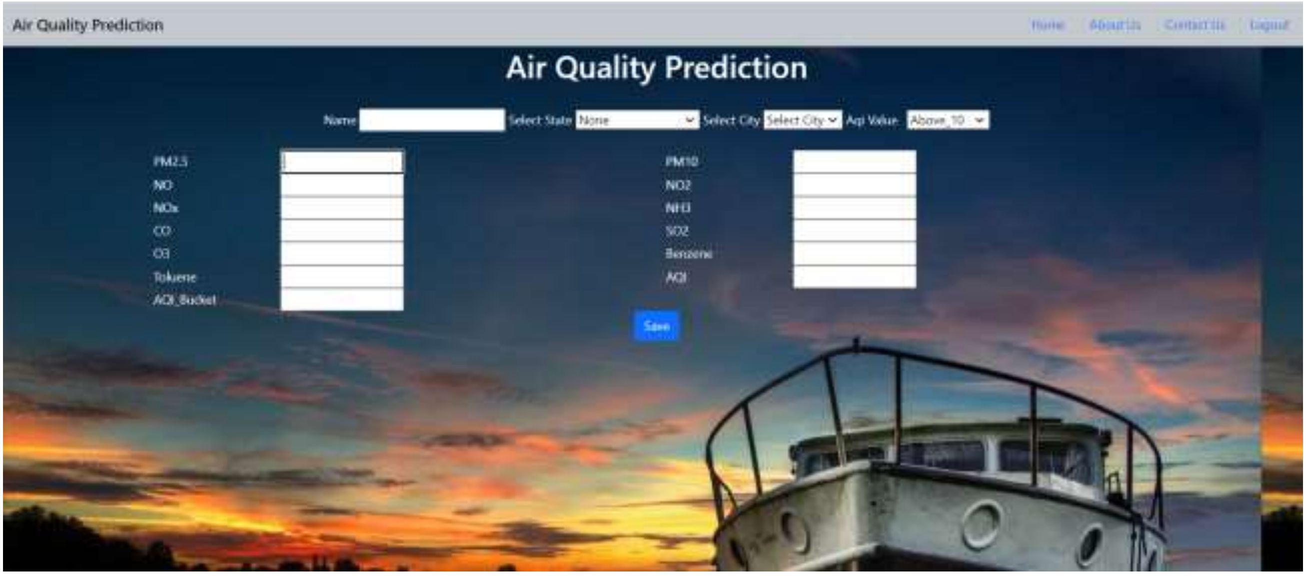


Fig.8.1 UI

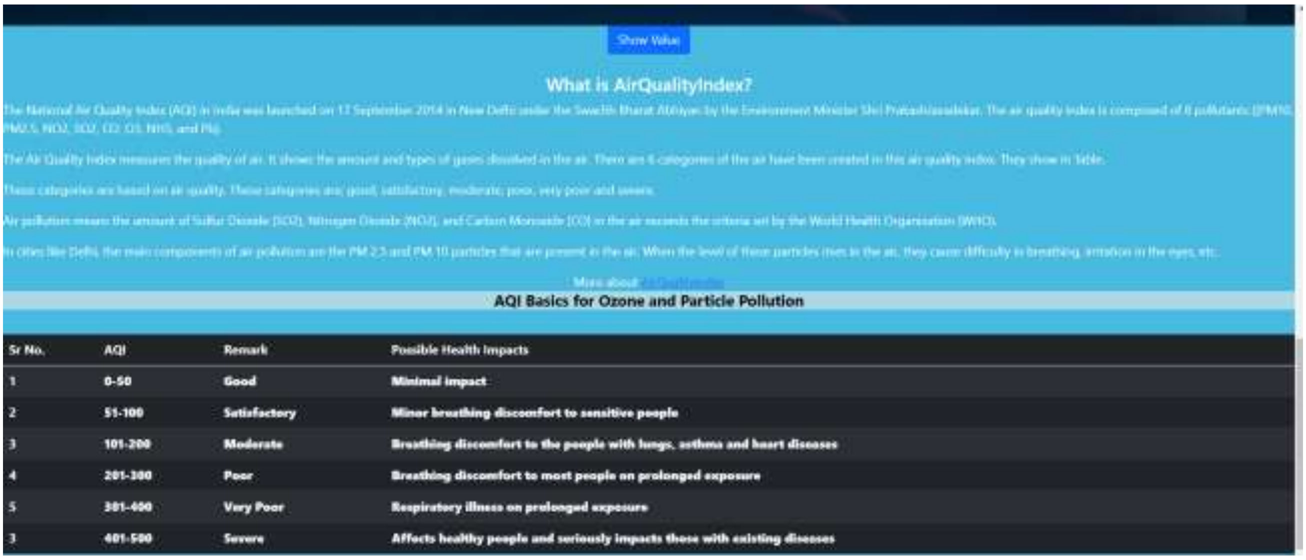


Fig.8.2 UI

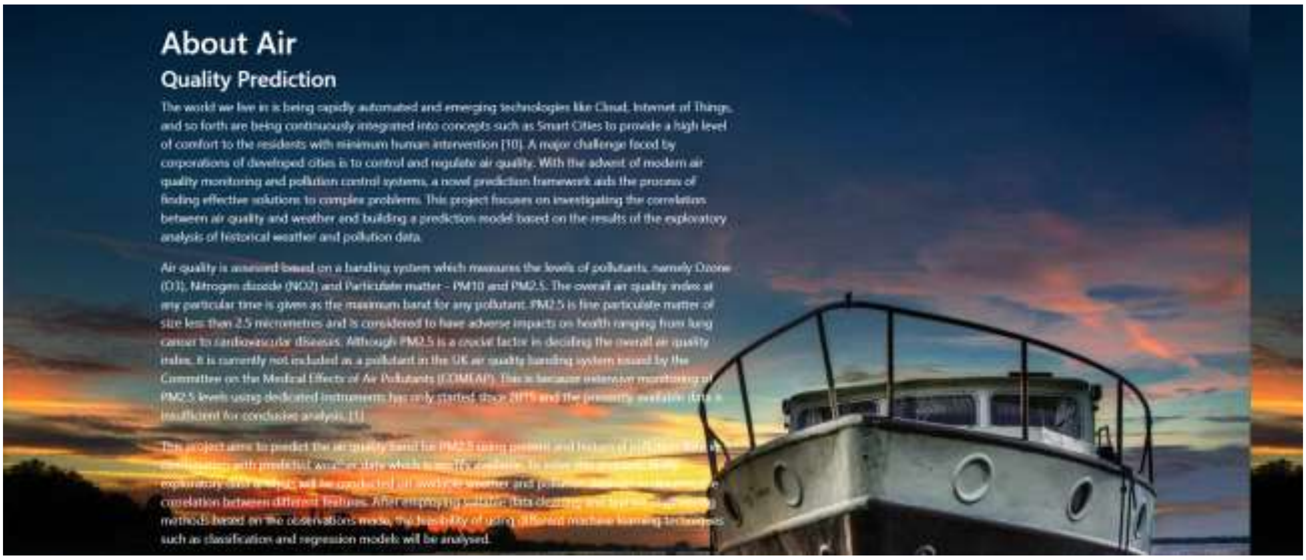


Fig.8.3 UI

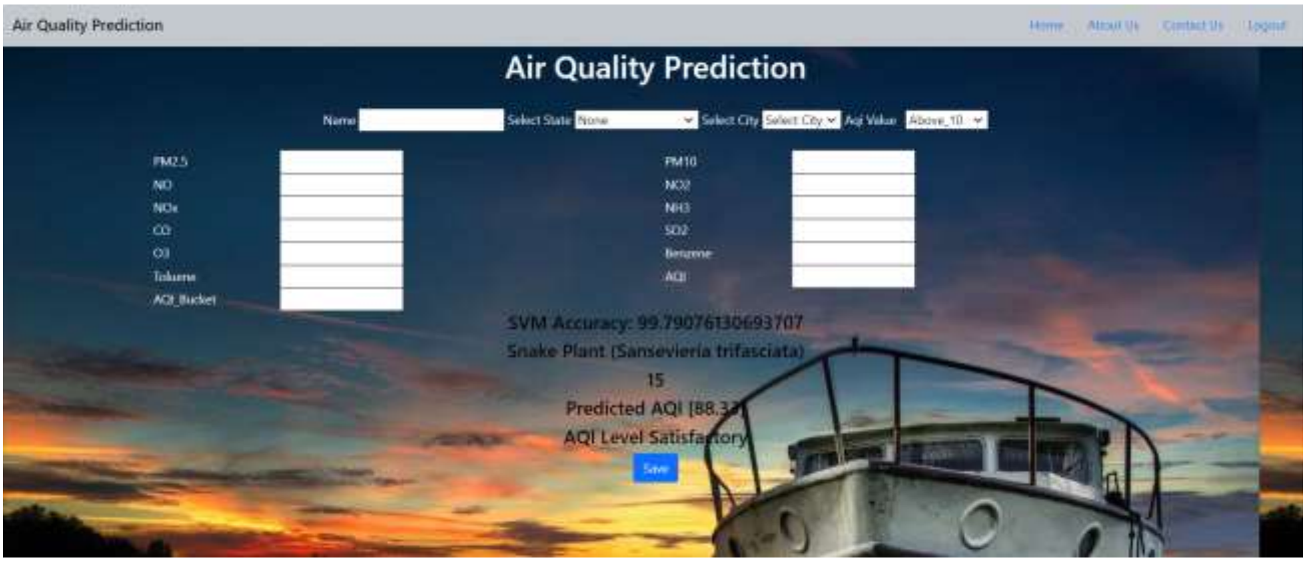


Fig.8.4 UI

# Application

- The main use of the **Air Quality Index** is to provide warnings to citizens, keeping them informed of the current conditions.
- This is particularly important for sensitive groups such as the elderly, children, or people with respiratory or heart problems. Such people should stay indoors when **air quality** is low.

# Chapter 6:Conclusion and Future scope

## Conclusion

1. AQI monitors provide an hourly and a daily average of Air pollution level of the specific areas.
2. This would be able to raise awareness of the level of pollution on the ground- level to the people who cause such pollution in the first place through their cars, back-door fires and air-conditioners etc.
3. An increasing range of adverse health effects has been linked to **air pollution**, and at even-lower concentrations.
4. The **Air Quality Index** (AQI) is a new public information tool that helps protect public health on a daily basis from the negative effects of **air pollution**.

## Future Work

- In the future we plan to analyse correlations between sensing sites located close to each other to cover latent similarities in pollutant or AQI patterns and to analyse If they are influenced by other environment factors such as green cover or traffic.
- We also plan to further extend the analysis of impact on air quality from different types of sensing areas across different cities.
- Another future pollution data patterns in different parts of a city according to its built environment.

# Chapter 7:References

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